question of why our universe has the properties it does. In plain language, things are the way they are in our universe because if they weren't, we wouldn't be here to notice. Like the winners of a mass game of Russian roulette, whose surprise at surviving is tempered by the realization that had they not won, they wouldn't have been able not to feel surprised, the multiverse hypothesis has the capacity to lessen our insistence on explaining why our universe appears as it does.

This line of argument is a version of an idea with a long history known as the anthropic principle. As presented, it is a perspective that is diametrically opposed to the dream of a rigid, fully predictive, unified theory in which things are the way they are because the universe could not be otherwise. Rather than being the epitome of poetic grace in which everything fits together with inflexible elegance, the multiverse and the anthropic principle paint a picture of a wildly excessive collection of universes with an insatiable appetite for variety. It will be extremely hard, if not impossible, for us ever to know if the multiverse picture is true. Even if there are other universes, we can imagine that we will never come into contact with any of them. But by vastly increasing the scope of "what's out there"—in a manner that dwarfs Hubble's realization that the Milky Way is but one galaxy among many—the concept of the multiverse does at least alert us to the possibility that we may be asking too much of an ultimate theory.

We should require that our ultimate theory give a quantum-mechanically consistent description of all forces and all matter. We should require that our ultimate theory give a cogent cosmology within our universe. However, if the multiverse picture is correct—a huge if—it may be asking too much for our theory to explain, as well, the detailed properties of the particle masses, charges, and the force strengths.

But we must emphasize that even if we accept the speculative premise of the multiverse, the conclusion that this compromises our predictive power is far from airtight. The reason, simply put, is that if we unleash our imaginations and allow ourselves to contemplate a multiverse, we should also unleash our theoretical musings and contemplate ways in which the apparent randomness of the multiverse can be tamed. For one relatively conservative musing, we can imagine that—were the multiverse picture true—we would be able to extend